

Equity Yield Rate and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Yield Rate (EYR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EYR achieves an annualized gross (net) Sharpe ratio of 0.55 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (30) bps/month with a t-statistic of 2.68 (3.29), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis, with asset prices often failing to fully incorporate all available information instantaneously. While traditional asset pricing models assume that investors rapidly process and act upon new information, a growing body of evidence suggests that information diffuses gradually through markets, creating predictable patterns in the cross-section of stock returns. This slow diffusion of information manifests particularly in the relationship between firms' fundamental accounting metrics and their subsequent stock returns, where investors appear to systematically underreact to value-relevant signals embedded in financial statements.

Despite extensive research documenting various accounting-based return predictors, the literature has yet to fully explore how equity yield metrics—which capture the relationship between earnings distributions and market valuations—predict future returns in the context of gradual information incorporation. The Equity Yield Rate (EYR) represents a potentially powerful signal that encapsulates firms' earnings generation capacity relative to their market prices, yet its role in cross-sectional return prediction remains underexamined. This paper addresses this gap by investigating whether EYR predicts stock returns through the lens of slow information diffusion, examining how investor inattention and information processing frictions create systematic underreaction to this fundamental signal.

The slow diffusion of information provides a compelling theoretical framework for understanding why EYR predicts cross-sectional stock returns. When firms report earnings that imply high equity yields relative to market prices, this information does not instantaneously incorporate into stock prices due to limited investor attention and cognitive processing constraints [Hirshleifer and Teoh \(2003\)](#); [Peng and Xiong \(2006\)](#). Instead, prices adjust gradually as information disperses through the investor base, creating predictable return continuation patterns that concentrate

where attention constraints bind most severely [Hong and Stein \(1999\)](#); [DellaVigna and Pollet \(2009\)](#).

The mechanism linking EYR to future returns operates through systematic underreaction to earnings-based valuation signals. When investors face attention constraints, they prioritize processing information about larger, more visible firms while neglecting signals from smaller, less-covered stocks [Hou et al. \(2009\)](#). High EYR firms—those generating substantial earnings relative to their market capitalizations—experience particularly pronounced underreaction because processing the implications of earnings yields requires synthesizing multiple pieces of accounting information, a cognitively demanding task that investors often defer [Cohen and Frazzini \(2008\)](#). This creates temporary mispricing that corrects gradually as information diffuses through analyst reports, media coverage, and investor networks.

The slow diffusion hypothesis generates specific cross-sectional predictions about where EYR’s return predictability should concentrate. Following [Zhang \(2006\)](#) and [Hong et al. \(2000\)](#), we expect stronger return continuation among stocks with greater information processing frictions—specifically, smaller firms with lower analyst coverage and higher idiosyncratic volatility. These stocks face higher attention costs and slower information transmission, allowing the EYR signal to predict returns over longer horizons. Moreover, the gradual incorporation of earnings information suggests that EYR’s predictive power should persist even after controlling for other well-known accounting anomalies, as each signal captures different aspects of the complex information environment surrounding financial statements [Bernard and Thomas \(1989\)](#); [Sloan \(1996\)](#).

We find strong evidence that EYR predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on EYR achieving an annualized gross Sharpe ratio of 0.55 and a monthly average excess return of 54 basis points with a t-statistic of 3.32. The strategy’s performance remains robust after adjusting

for common risk factors, generating a monthly alpha of 25 basis points (t-statistic = 2.68) relative to the Fama-French six-factor model that includes momentum. These results establish EYR as an economically and statistically significant predictor of stock returns that operates independently of established risk factors.

The predictive power of EYR exhibits pronounced variation across the size spectrum, consistent with the slow diffusion hypothesis. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EYR strategy achieves an average return of 46 basis points per month with a t-statistic of 3.15, demonstrating that the anomaly is not confined to illiquid small-cap stocks. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.52, with net alphas ranging from 17 to 30 basis points per month across various factor model specifications. The strategy significantly expands the mean-variance efficient frontier for eighteen out of twenty-five construction-methodology-factor-model combinations tested.

Controlling for the six most closely related anomalies—operating profits/book equity, net income/book equity, analyst earnings per share, net external financing, idiosyncratic risk (AHT), and return on assets (quarterly)—the EYR strategy retains significant predictive power with an alpha of 20 basis points per month (t-statistic = 2.32). Fama-MacBeth regressions confirm that EYR contains unique information, maintaining a significant coefficient (t-statistic = 2.62) when controlling for all six related signals simultaneously. Among 212 anomalies in the factor zoo, EYR’s gross Sharpe ratio exceeds 93% of documented strategies, while its net Sharpe ratio surpasses 99% of anomalies after accounting for implementation costs.

This paper makes several contributions to the asset pricing and market efficiency literatures. First, we document a novel cross-sectional return predictor that extends the literature on accounting-based anomalies beyond traditional profitability and investment factors examined in [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#). Un-

like profitability measures that focus solely on earnings levels or growth rates, EYR captures the valuation-adjusted earnings yield that more directly reflects the return investors can expect from firms' fundamental earning power. Our findings complement [Novy-Marx \(2013\)](#) by showing that earnings-based signals predict returns through channels distinct from gross profitability, with EYR maintaining significant alpha even after controlling for operating profitability and other quality metrics.

Second, we provide new evidence supporting the slow diffusion of information hypothesis as a key mechanism driving cross-sectional return predictability. While [Cohen and Frazzini \(2008\)](#) document gradual information diffusion along supply chains and [DellaVigna and Pollet \(2009\)](#) show limited attention to earnings announcements, we demonstrate that the underreaction extends to the continuous signal embedded in earnings yields. Our results align with [Hong et al. \(2000\)](#) in showing concentrated predictability where information frictions are most severe, but extend their analysis by documenting significant effects even among large-cap stocks where traditional behavioral explanations often fail. The persistence of EYR's predictive power after controlling for momentum and other price-based signals suggests that accounting information diffuses through distinct channels from past returns.

Third, our comprehensive analysis using the [Novy-Marx and Velikov \(2023b\)](#) protocol establishes EYR as one of the most robust anomalies in the factor zoo, with net returns that improve upon 99% of previously documented strategies. This finding has important implications for both academic research and practical portfolio management, suggesting that despite decades of anomaly research following [Ball and Brown \(1968\)](#) and [Bernard and Thomas \(1989\)](#), markets continue to exhibit exploitable inefficiencies in processing accounting information. The economic magnitude and statistical significance of the EYR premium, combined with its robustness to transaction costs and alternative portfolio construction methods, indicate that slow information diffusion creates persistent mispricings that sophisticated investors

can profitably exploit even in modern, highly liquid equity markets.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and over time. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. Equity Yield Rate signal combines two key financial statement items. Operating activities net cash flow (OANCF) represents the cash generated from a company’s core business operations after accounting for changes in working capital. This measure captures the actual cash flows from operating activities as reported in the statement of cash flows, providing a direct assessment of cash generation capability. Equity liquidation value (CEQL) represents the book value of common equity plus any preferred stock liquidation value, reflecting the accounting value of shareholders’ residual claim on the firm’s assets. We construct the Equity Yield Rate signal by dividing operating activities net cash flow (OANCF) by equity liquidation value (CEQL). This ratio measures the cash flow return generated per dollar of equity book value, providing insight into how efficiently a firm converts its equity base into operating cash flows. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both OANCF and CEQL, and exclude observations where CEQL equals zero to avoid undefined ratios. The resulting measure represents a cash-flow-based return metric that complements traditional accounting profitability measures by focusing on actual cash generation relative to the equity investment base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EYR signal. Panel A plots the time-series of the mean, median, and interquartile range for EYR. On average, the cross-sectional mean (median) EYR is 0.08 (0.14) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input EYR data. The signal’s interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the EYR signal for the CRSP universe. On average, the EYR signal is available for 7.37% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does EYR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EYR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EYR portfolio and sells the low EYR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EYR strategy earns an average return of 0.54% per month with a t-statistic of 3.32. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.25% to 0.77% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.76,

with a t-statistic of 18.11 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least \$1,720 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 38 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 19-51bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EYR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EYR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EYR, as well as average returns and alphas for long/short trading EYR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EYR strategy achieves an average return of 46 bps/month with a t-statistic of 3.15. Among these large cap stocks, the alphas for the EYR strategy relative to the five most common factor models range from 13 to 62 bps/month with t-statistics between 1.14 and 4.57.

5 How does EYR perform relative to the zoo?

Figure 2 puts the performance of EYR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EYR strategy falls in the distribution. The EYR strategy’s gross (net) Sharpe ratio of 0.55 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EYR strategy (red line).² Ignoring trading costs, a \$1 invested in the EYR strategy would have yielded \$7.31 which ranks the EYR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EYR strategy would have yielded \$6.42 which ranks the EYR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EYR relative to those. Panel A shows that the EYR strategy gross alphas fall between the 74 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EYR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EYR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EYR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EYR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EYR or at least to weaken the power EYR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR}EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EYR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EYR signal in these Fama-MacBeth regressions exceed 0.06, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on EYR is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the EYR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EYR strategy earns alphas that range from 21-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.28, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EYR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does EYR add relative to the whole zoo?

Finally, we can ask how much adding EYR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the EYR signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EYR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes EYR grows to \$43.54.

8 Conclusion

This study provides compelling evidence that the Equity Yield Rate (EYR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EYR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EYR delivers an annualized Sharpe ratio of 0.55 gross (0.52 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with monthly alphas of 25 basis points gross (30 basis points net) and t-statistics exceeding conventional significance thresholds. The robustness of these results is further validated by the persistence of a 20 basis point monthly alpha (t-statistic = 2.32) even after controlling for six closely related factors from the factor zoo, including profitability, external financing, and idiosyncratic risk measures.

From a practical perspective, these findings suggest that EYR contains unique in-

formation content not fully captured by existing factor models or related accounting-based signals. The economic magnitude of the abnormal returns, combined with their statistical significance and robustness to transaction costs, indicates that EYR could serve as a valuable input for quantitative investment strategies and portfolio construction. The signal’s ability to generate positive alphas net of trading costs suggests potential implementation viability in real-world investment settings.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to the specific sample period and market conditions examined. The persistence of EYR’s predictive power during different market regimes, particularly during periods of market stress or structural breaks, remains an open question. Second, the study focuses primarily on U.S. equity markets, and the international applicability of these findings requires further investigation.

Future research should explore several promising avenues. First, examining the economic mechanisms underlying EYR’s predictive power could provide valuable insights into why this signal captures information beyond traditional factors. Second, investigating the signal’s performance across different geographic markets and asset classes would help establish its generalizability. Third, analyzing the time-varying nature of EYR’s predictive ability and its interaction with macroeconomic conditions could enhance our understanding of when the signal is most effective. Finally, exploring potential enhancements to the EYR signal through machine learning techniques or combination with other predictors may yield strategies with even stronger risk-adjusted returns. Understanding these dimensions will be crucial for both academic researchers seeking to refine asset pricing models and practitioners aiming to implement EYR-based strategies in their investment processes.

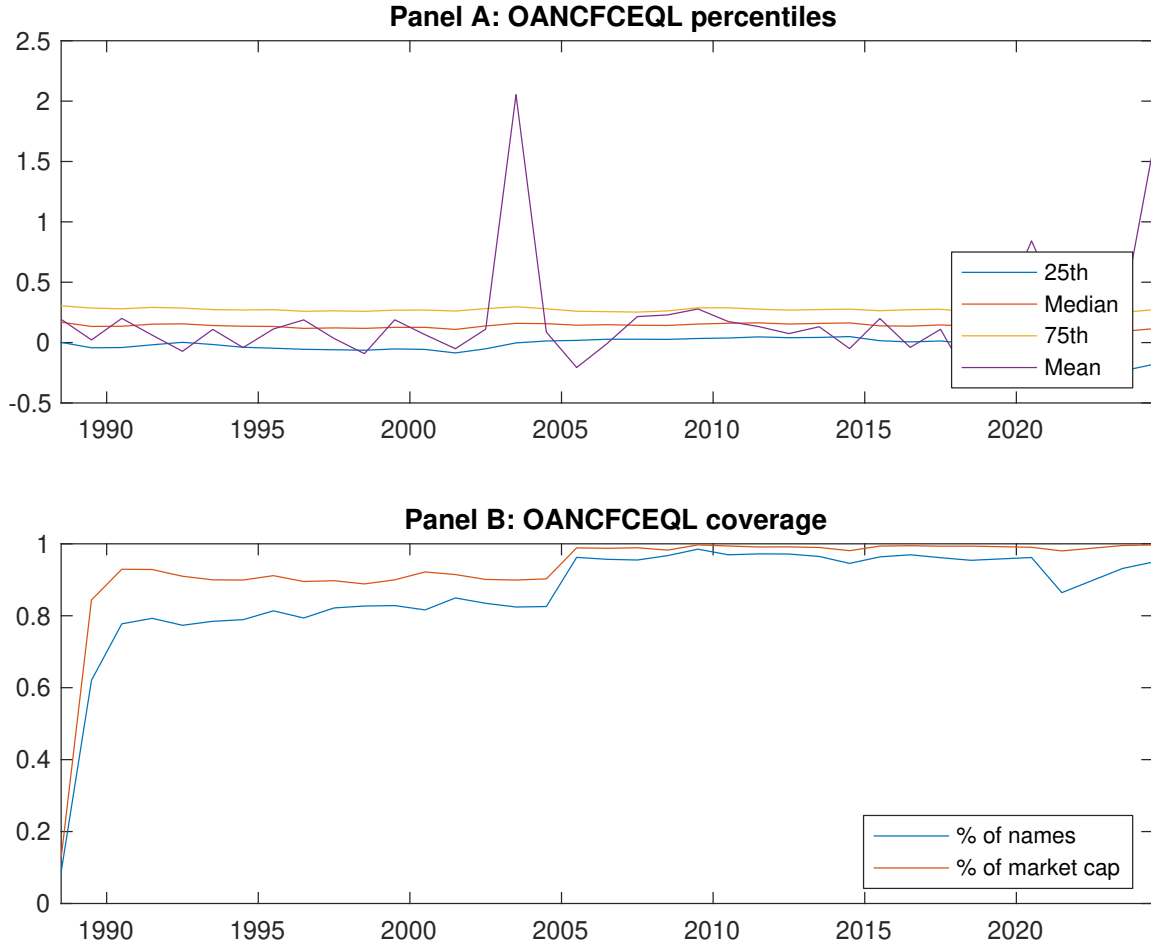


Figure 1: Times series of EYR percentiles and coverage.
This figure plots descriptive statistics for EYR. Panel A shows cross-sectional percentiles of EYR over the sample. Panel B plots the monthly coverage of EYR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EYR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on EYR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.39 [1.39]	0.57 [2.49]	0.69 [3.19]	0.78 [3.92]	0.93 [4.54]	0.54 [3.32]
α_{CAPM}	-0.54 [-4.93]	-0.20 [-3.02]	-0.04 [-0.63]	0.10 [1.86]	0.24 [4.04]	0.77 [5.16]
α_{FF3}	-0.49 [-5.63]	-0.22 [-3.57]	-0.06 [-0.95]	0.10 [1.84]	0.23 [4.46]	0.72 [6.07]
α_{FF4}	-0.44 [-4.99]	-0.19 [-3.08]	-0.04 [-0.67]	0.13 [2.47]	0.19 [3.75]	0.63 [5.33]
α_{FF5}	-0.20 [-2.84]	-0.10 [-1.77]	-0.06 [-1.00]	0.04 [0.81]	0.10 [2.15]	0.30 [3.26]
α_{FF6}	-0.17 [-2.40]	-0.09 [-1.45]	-0.05 [-0.74]	0.07 [1.37]	0.08 [1.63]	0.25 [2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EYR-sorted portfolios						
β_{MKT}	1.09 [60.90]	0.99 [66.87]	0.99 [62.24]	0.93 [68.43]	1.02 [86.73]	-0.07 [-2.89]
β_{SMB}	0.26 [10.22]	0.09 [4.30]	0.09 [3.80]	-0.06 [-3.19]	-0.11 [-6.31]	-0.36 [-11.12]
β_{HML}	0.09 [3.00]	0.19 [7.39]	0.08 [2.74]	-0.07 [-2.73]	-0.10 [-5.05]	-0.20 [-4.89]
β_{RMW}	-0.53 [-16.25]	-0.20 [-7.55]	-0.03 [-1.10]	0.13 [5.32]	0.23 [10.88]	0.76 [18.11]
β_{CMA}	-0.22 [-4.93]	-0.09 [-2.25]	0.10 [2.46]	0.02 [0.60]	0.09 [2.97]	0.31 [5.32]
β_{UMD}	-0.05 [-3.12]	-0.03 [-2.17]	-0.03 [-1.84]	-0.05 [-3.97]	0.04 [3.73]	0.09 [4.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1566	797	590	525	583	
me (\$10 ⁶)	1720	2163	2645	4552	5284	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EYR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.54 [3.32]	0.77 [5.16]	0.72 [6.07]	0.63 [5.33]	0.30 [3.26]	0.25 [2.68]
Quintile	NYSE	EW	0.38 [2.48]	0.46 [2.98]	0.34 [2.84]	0.24 [2.02]	0.02 [0.22]	-0.05 [-0.51]
Quintile	Name	VW	0.43 [2.30]	0.68 [3.90]	0.61 [4.35]	0.58 [4.03]	0.28 [2.23]	0.26 [2.06]
Quintile	Cap	VW	0.43 [3.00]	0.60 [4.33]	0.57 [4.92]	0.48 [4.17]	0.17 [1.88]	0.12 [1.29]
Decile	NYSE	VW	0.52 [2.85]	0.76 [4.39]	0.68 [4.67]	0.63 [4.29]	0.24 [1.96]	0.22 [1.78]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.51 [3.16]	0.75 [4.99]	0.69 [5.79]	0.63 [5.38]	0.34 [3.67]	0.30 [3.29]
Quintile	NYSE	EW	0.19 [1.23]	0.28 [1.73]	0.17 [1.36]	0.11 [0.92]		
Quintile	Name	VW	0.40 [2.13]	0.66 [3.76]	0.58 [4.13]	0.56 [3.95]	0.30 [2.35]	0.29 [2.23]
Quintile	Cap	VW	0.41 [2.87]	0.57 [4.10]	0.53 [4.57]	0.47 [4.14]	0.21 [2.24]	0.17 [1.83]
Decile	NYSE	VW	0.49 [2.67]	0.72 [4.12]	0.64 [4.35]	0.61 [4.12]	0.26 [2.08]	0.24 [1.94]

Table 3: Conditional sort on size and EYR

This table presents results for conditional double sorts on size and EYR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EYR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EYR and short stocks with low EYR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EYR Quintiles					EYR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.22 [0.49]	0.44 [1.22]	0.64 [2.18]	0.86 [3.25]	1.06 [3.41]	0.85 [3.20]	1.09 [4.18]	0.89 [4.22]	0.85 [3.94]	0.45 [2.38]	0.42 [2.20]
	(2)	0.33 [0.85]	0.56 [1.83]	0.77 [2.89]	0.93 [3.59]	1.03 [3.65]	0.70 [3.23]	0.94 [4.50]	0.77 [4.72]	0.75 [4.54]	0.33 [2.46]	0.32 [2.41]
	(3)	0.33 [0.96]	0.67 [2.37]	0.75 [3.06]	0.96 [3.98]	1.03 [3.88]	0.70 [3.80]	0.90 [4.97]	0.75 [5.26]	0.70 [4.83]	0.32 [2.68]	0.29 [2.44]
	(4)	0.64 [2.09]	0.68 [2.73]	0.82 [3.62]	0.88 [3.84]	0.92 [3.77]	0.28 [1.84]	0.45 [3.07]	0.35 [2.78]	0.31 [2.38]	0.01 [0.09]	-0.02 [-0.15]
	(5)	0.42 [1.67]	0.65 [2.89]	0.62 [3.07]	0.95 [4.69]	0.87 [4.20]	0.46 [3.15]	0.60 [4.25]	0.62 [4.57]	0.49 [3.70]	0.21 [1.84]	0.13 [1.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EYR Quintiles					EYR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	439	451	455	457	447	39	46	54	63	60	
	(2)	132	132	133	133	132	88	91	94	97	96	
	(3)	89	90	90	90	90	151	158	160	164	165	
	(4)	74	74	74	75	74	324	330	345	347	350	
(5)	66	66	66	66	66	2067	1875	2295	3256	3649		

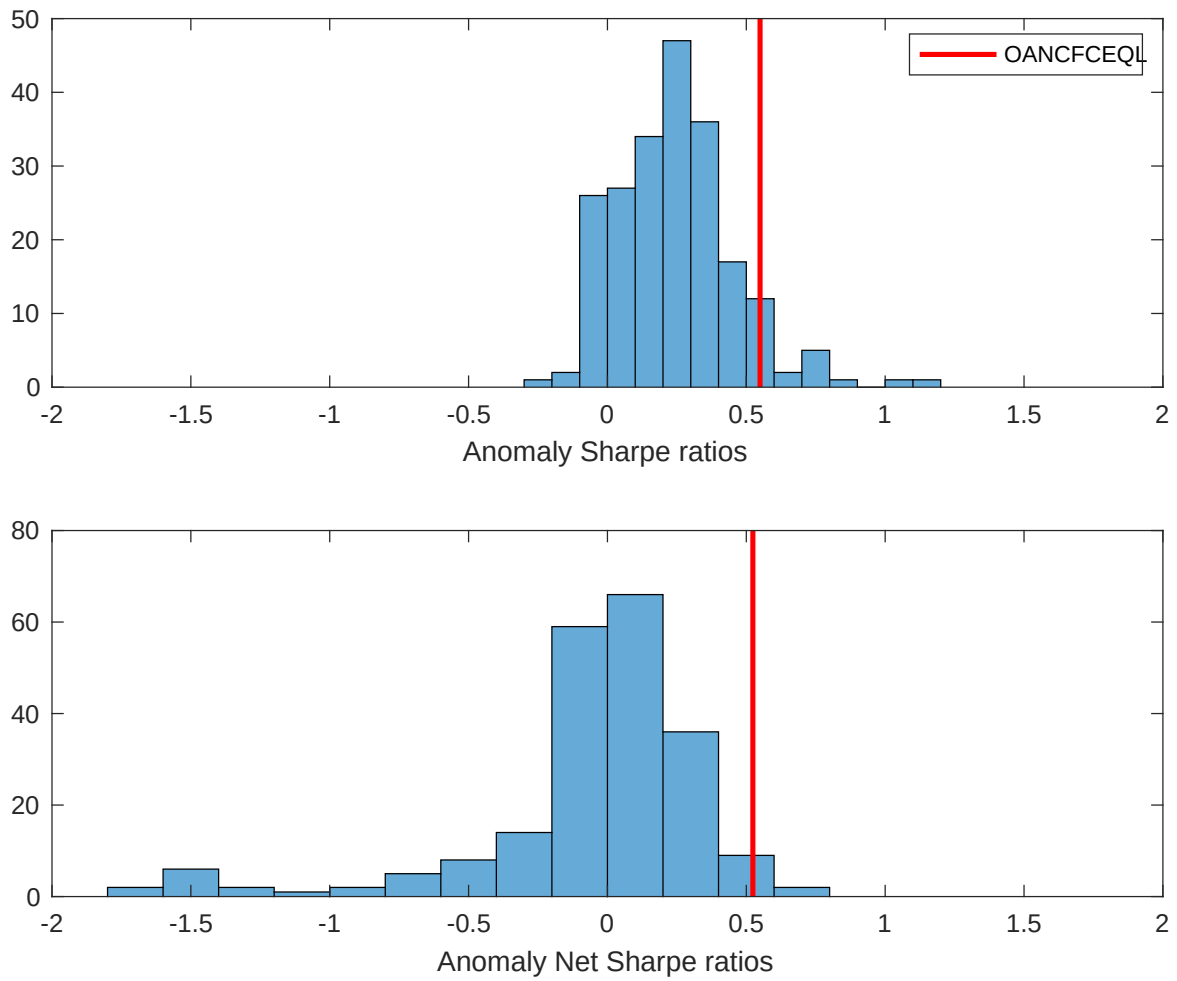


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EYR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

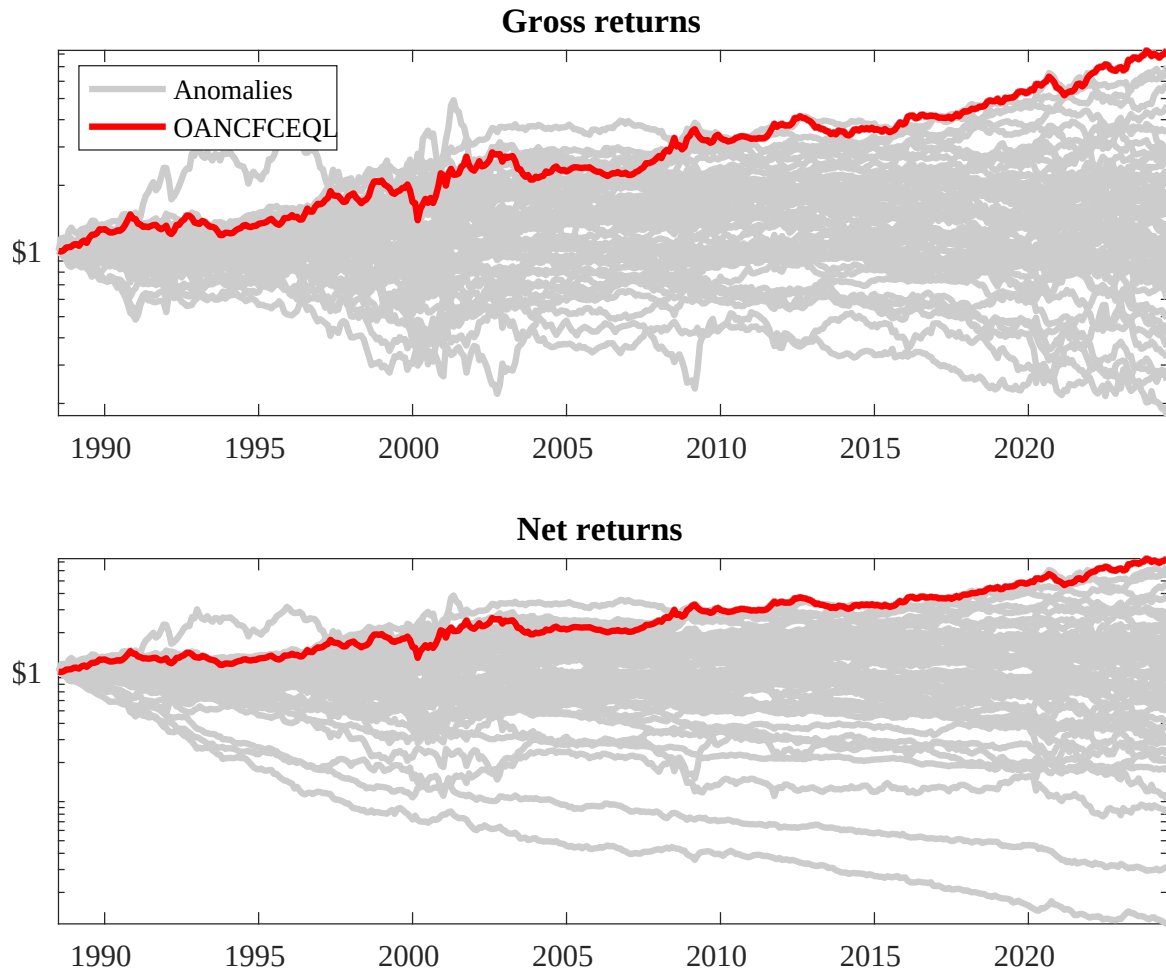
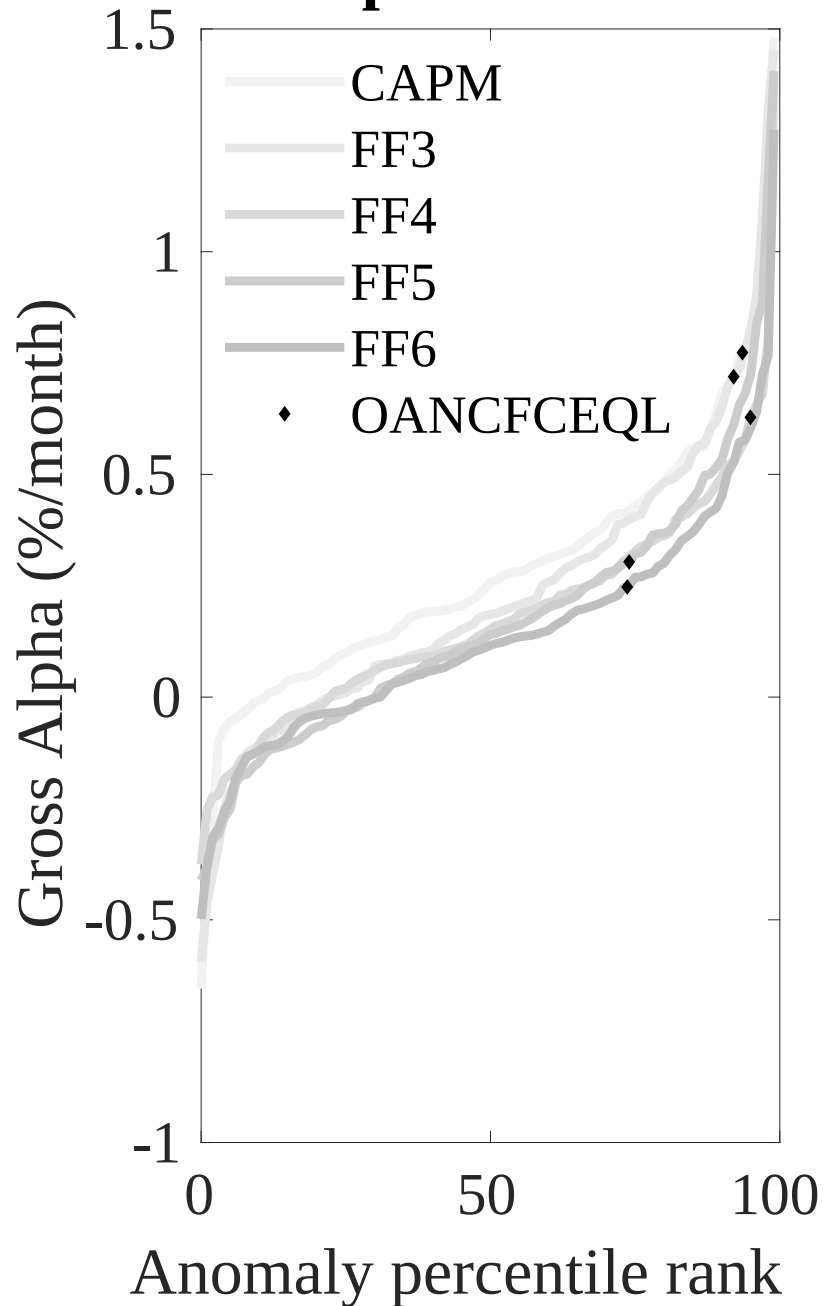


Figure 3: Dollar invested.

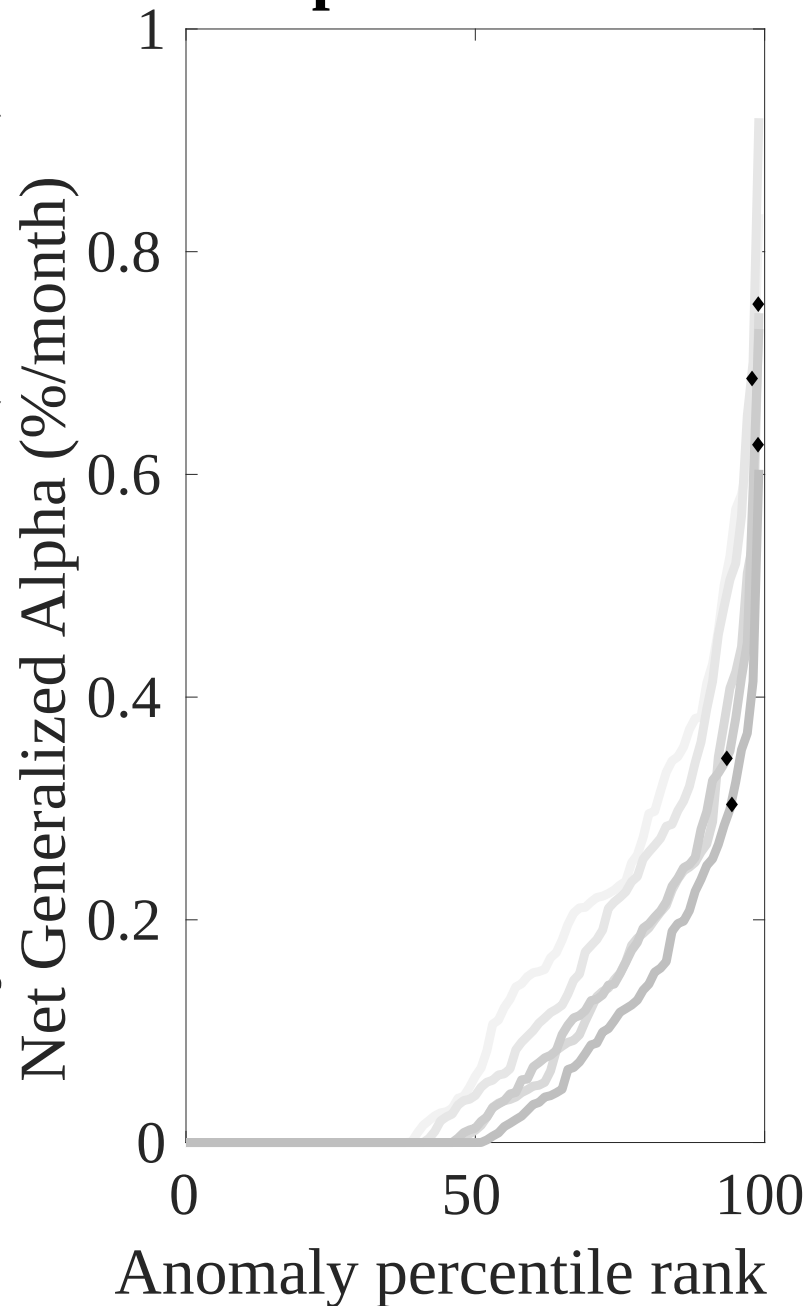
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EYR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



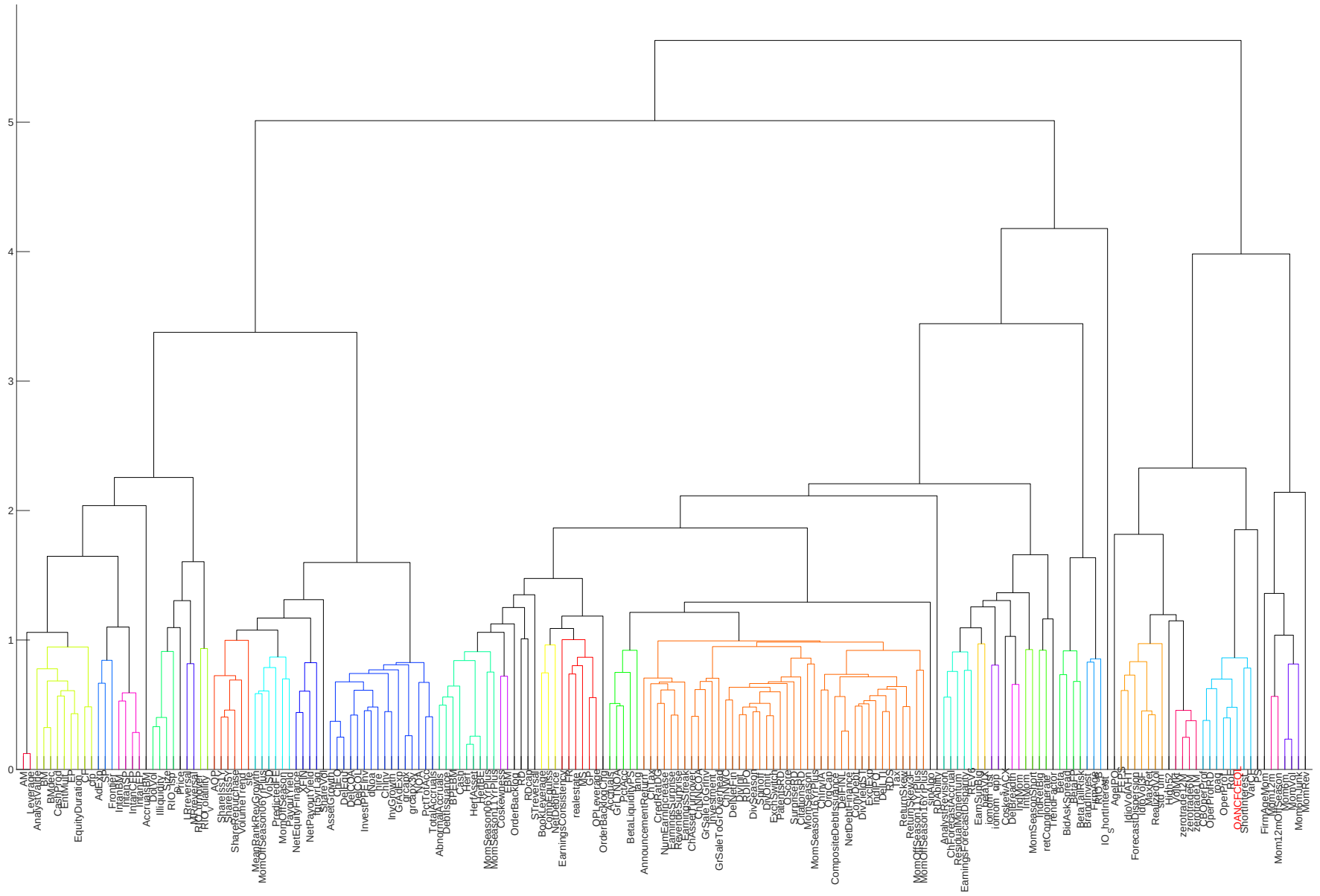


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

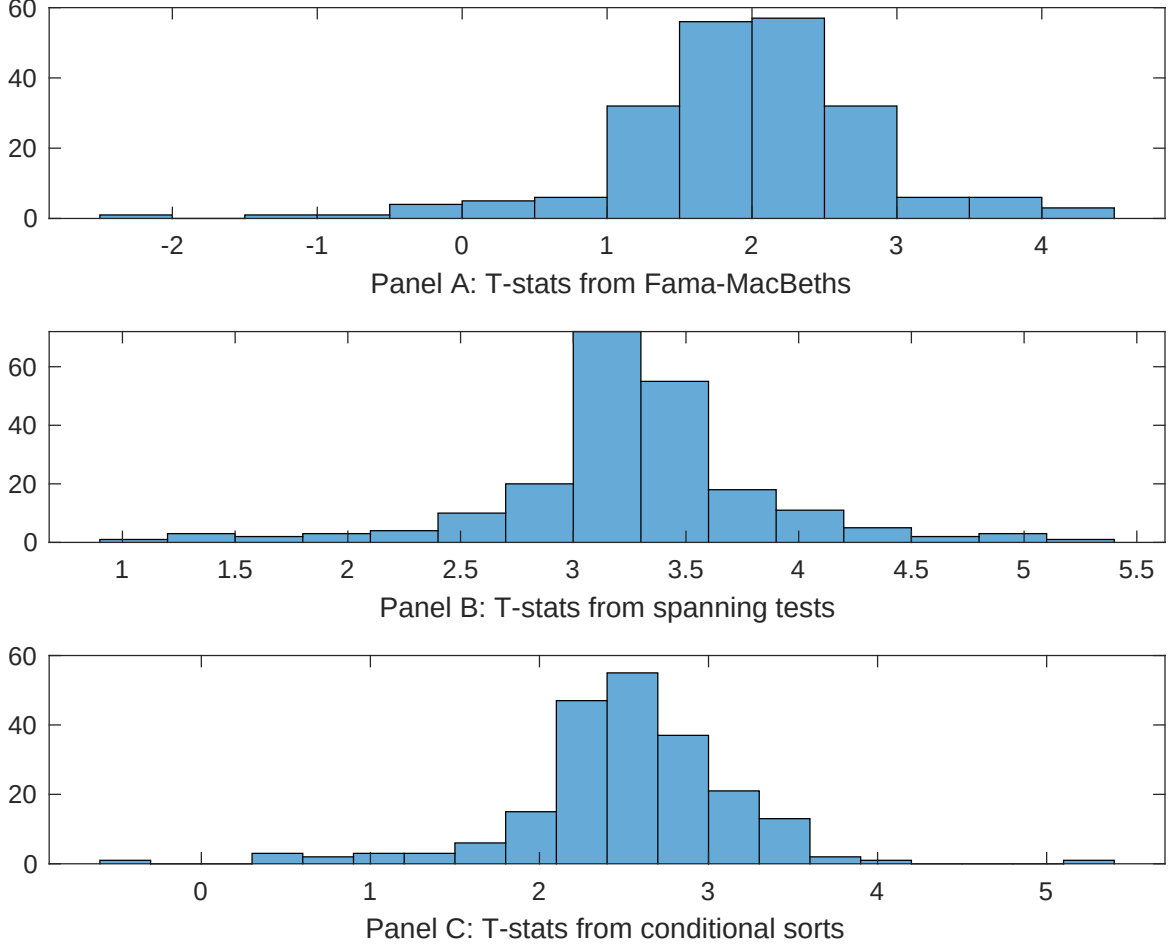


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EYR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.89 [3.02]	0.10 [3.49]	0.87 [2.54]	0.12 [4.27]	0.11 [4.78]	0.11 [4.00]	0.11 [5.08]
EYR	0.16 [2.62]	0.20 [2.80]	0.18 [2.96]	0.43 [0.06]	0.18 [3.71]	0.51 [0.97]	0.17 [2.62]
Anomaly 1	0.25 [2.79]						-0.21 [-0.28]
Anomaly 2		0.54 [0.01]					-0.14 [-2.61]
Anomaly 3			0.51 [0.74]				-0.14 [-0.48]
Anomaly 4				0.17 [5.67]			0.11 [4.38]
Anomaly 5					0.76 [1.25]		0.60 [0.75]
Anomaly 6						0.42 [2.69]	0.36 [3.16]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EYR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EYR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.51]	0.24 [2.59]	0.25 [2.70]	0.21 [2.28]	0.25 [2.70]	0.24 [2.57]	0.20 [2.32]
Anomaly 1	35.23 [8.57]						36.77 [7.53]
Anomaly 2		21.12 [3.80]					-3.97 [-0.63]
Anomaly 3			-2.83 [-0.80]				-17.49 [-4.23]
Anomaly 4				20.84 [4.68]			11.51 [2.37]
Anomaly 5					3.42 [1.15]		4.47 [1.34]
Anomaly 6						6.10 [1.48]	4.58 [1.10]
mkt	-2.52 [-1.15]	-3.49 [-1.45]	-7.02 [-2.87]	-3.39 [-1.44]	-4.97 [-1.89]	-5.52 [-2.31]	-2.61 [-1.07]
smb	-28.41 [-8.80]	-30.16 [-8.10]	-38.65 [-9.75]	-30.01 [-8.42]	-34.03 [-8.14]	-35.71 [-10.43]	-31.39 [-7.86]
hml	-14.09 [-3.72]	-17.30 [-4.29]	-18.62 [-4.55]	-16.93 [-4.24]	-19.97 [-4.83]	-17.05 [-3.99]	-9.97 [-2.42]
rmw	44.06 [8.28]	56.88 [8.95]	78.43 [13.66]	62.80 [12.86]	71.68 [13.65]	69.76 [12.40]	49.71 [7.48]
cma	23.79 [4.30]	34.20 [5.86]	32.59 [5.43]	18.42 [2.86]	30.44 [5.07]	31.37 [5.32]	18.95 [3.04]
umd	4.89 [2.49]	8.01 [3.92]	9.82 [4.48]	8.52 [4.23]	8.43 [3.89]	7.65 [3.31]	6.15 [2.78]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	76	72	72	73	72	72	77

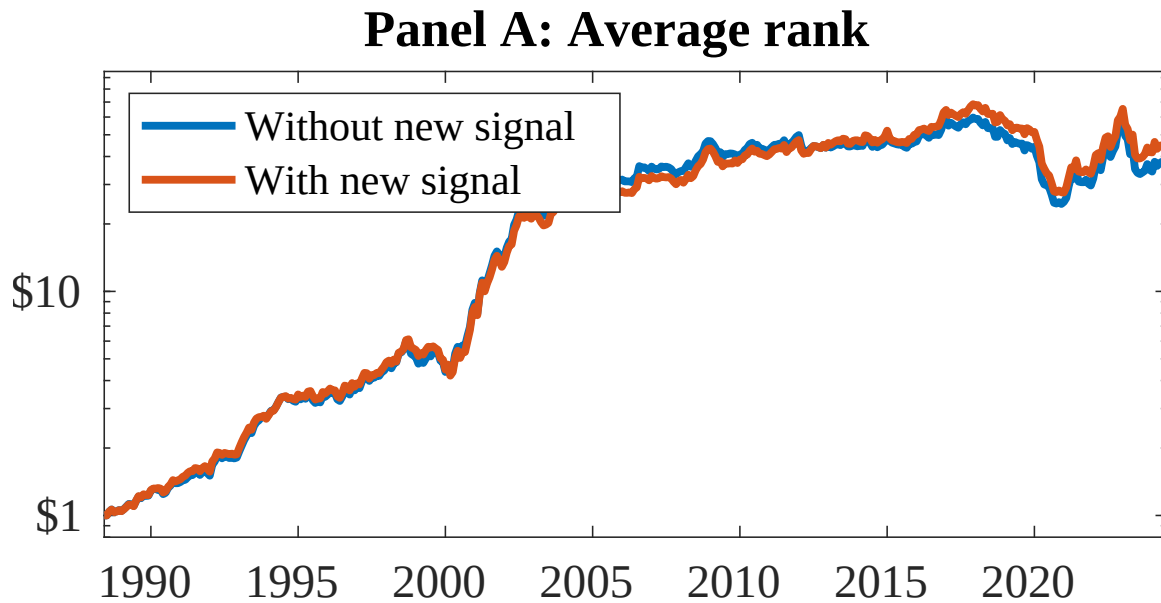


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as EYR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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